

1. A network shows an open circuit voltage of 9.0V and a short-circuit current of 1.5A at its terminals. Find R_{th} & the terminal voltage for a 4Ω load.

Ans) Open circuit voltage, $V_{th} = 9\text{V}$
 Short circuit current, $I_{sc} = I_{N} = 1.5\text{A}$
 Load Resistance, $R_L = 4\Omega$

To find R_{th} ,

From Thevenin - Norton relationship -

$$R_{th} = \frac{V_{th}}{I_N} = \frac{9}{1.5} = 6\Omega \quad \therefore R_{th} = R_N = 6\Omega$$

Terminal voltage for 4Ω load -

Using Thevenin's equivalent ckt -

$$V_L = V_{th} \frac{R_L}{R_{th} + R_L} = 9 \times \frac{4}{6+4} = 3.6\text{V}$$

$$\therefore V_L = 3.6\text{V}$$

2. A 100mH inductor & a $47\mu\text{F}$ capacitor are placed in series. Find the frequency at which their reactances are equal.

Ans) Given, $L = 100\text{mH} = 0.1\text{H}$
 $C = 47\mu\text{F} = 47 \times 10^{-6}\text{F}$

at equal reactance,

$$X_L = X_C$$

$$\Rightarrow \omega L = \frac{1}{\omega C}$$

$$\therefore \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.1)(47 \times 10^{-6})}} = 461.9\text{ rad/s}$$

$$\text{now, } f_0 = \frac{\omega_0}{2\pi} = \frac{461.9}{2\pi} = 73.51 \text{ Hz}$$

3. An inductor voltage waveform is $+20\text{V}$ for 3ms & $-V_2$ for the remaining 2ms of a 5ms period. Find V_2 .

Ans) Given,

$$\text{Total period} = 5\text{ms}$$

$$\text{at steady state operations, } \int_0^T V_L(t) dt = 0$$

$\therefore$ Positive & negative volt-seconds must be equal.

Apply volt-second balance

$$(20)(3 \times 10^{-3}) + (-V_2)(2 \times 10^{-3}) = 0$$

$$\Rightarrow 60 \times 10^{-3} - 2V_2 \times 10^{-3} = 0$$

$$\Rightarrow 60 = 2V_2 \quad \Rightarrow V_2 = 30\text{V}$$

$\therefore$ waveform of an inductor $+20\text{V}$ for 3ms & -30V for 2ms .

4. Sketch the impedance locus of a series RLC circuit as ω is swept from 0 to ∞ & identify the sketch at which the locus cross the real axis.

Ans) For a series RLC circuit,

$$Z = R + jX_L + jX_C$$

$$\text{We know, } X_L = \omega L \text{ \& } X_C = \frac{1}{\omega C}$$

$$\therefore Z = R + j\left(\omega L + \frac{1}{\omega C}\right)$$

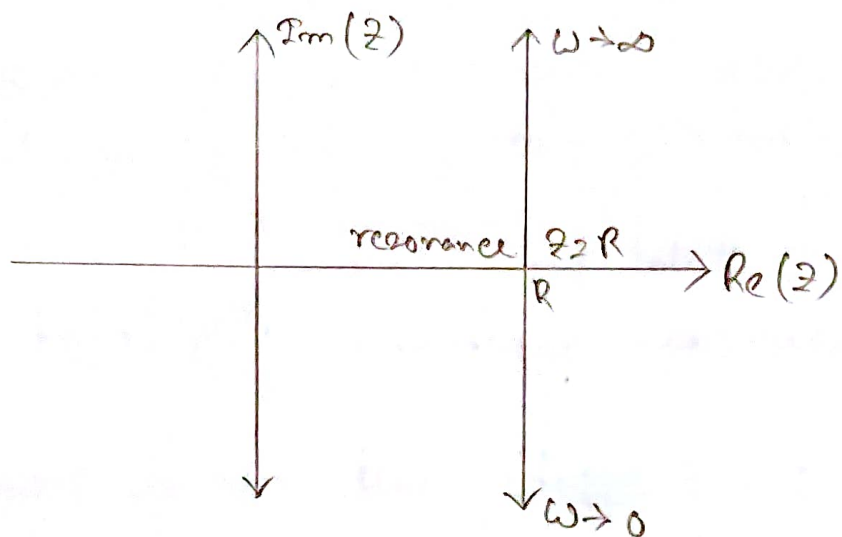
$$\therefore \text{Re}(Z) = R \text{ \& } \text{Im}(Z) = \omega L - \frac{1}{\omega C}$$

$$\text{at } \omega = 0, X_L = 0 \text{ but } X_C \rightarrow \infty \therefore \text{Im}(Z) \rightarrow -\infty$$

$\therefore$ The locus starts at $R - j\infty$

Impedance locus,

The locus is a vertical straight line at $\text{Re}(z) = R$:



at resonance,

The locus crosses the real axis when $x_L = x_C$

$$\Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

at this point, $x_L = x_C = 0$

$$\Rightarrow Z = R$$

at $\omega \rightarrow \infty$, $x_L \rightarrow \infty$ & $x_C \rightarrow 0$

$$\Rightarrow \text{Im}(Z) \rightarrow +\infty$$

so the locus goes toward $R + j\infty$.

5. Using Euler's Formula, express $v(t) = 20 \cos(314t + 30^\circ)$ as a phasor, & state the frequency in Hz.

Ans) Given,

$$v(t) = 20 \cos(314t + 30^\circ)$$

std. sinusoidal waveform - $v(t) = V_m \cos(\omega t + \phi)$

here, $V_m = 20$, $\omega = 314 \text{ rad/s}$, $\phi = 30^\circ$

$\Rightarrow$ Phasor is $20 \angle 30^\circ \text{ V} \Rightarrow V'$

$$V_{\text{rms}} = \frac{20}{\sqrt{2}} = 14.14 \text{ V} \quad \text{Hence, } V' = 14.14 \angle 30^\circ \text{ V}_{\text{rms}}$$

We know,

$$\omega = 2\pi f \Rightarrow f = 314/2\pi$$

$$= 49.97 \text{ Hz}$$

$$\Rightarrow f \approx 50 \text{ Hz}$$